

Problem Set 11 Answers

1. Let X_1 be Alice's result and X_2 be Bob's result. Then we have $X = X_1 + X_2$ so $E(X) = E(X_1 + X_2)$ but expectation is linear, so $E(X) = E(X_1) + E(X_2)$. We have $E(X_1) = (1/6)(1+2+3+4+5+6) = 21/6 = 7/2$ and $E(X_2) = (1/20)(1+2+3+\cdots+20) = 210/20 = 21/2$, so we conclude that $E(X) = 7/2 + 21/2 = 14$.

2. We have $E(X_1) = 1/6$ and $E(X_2) = 1/6$ and $E(X_3) = 1/6$. We also have $E(X) = E(X_1 + X_2 + X_3)$, and by linearity of expectation, it follows that $E(X) = 1/6 + 1/6 + 1/6 = 1/2$.

3. We note that $X_j = 1$ if and only if the j th labelled Purdue shirt is chosen before the first IU t-shirt (and therefore before all of the IU t-shirts), so $P(X_j = 1) = 1/4$, and thus $E(X_j) = 1/4$. Since we already noted that $X = 1 + X_1 + X_2 + \cdots + X_7$, it follows by linearity of expectation that $E(X) = E(1 + X_1 + X_2 + \cdots + X_7) = 1 + 1/4 + \cdots + 1/4 = 1 + 7/4 = 11/4$.

4. We have $E(X_j) = 1/3$ for each j . We also have $E(X) = E(X_1 + X_2 + X_3 + X_4 + X_5)$, and by linearity of expectation, it follows that $E(X) = 1/3 + 1/3 + 1/3 + 1/3 + 1/3 = 5/3$.

5. We compute $E(U) = \sum_{u=0}^{\infty} (u) \left(\frac{e^{-1}}{u!} \right)$. The $u = 0$ term is 0, and for $u \geq 1$, we can simplify $u/u! = 1/(u-1)!$, so we get $E(U) = \sum_{u=1}^{\infty} \frac{e^{-1}}{(u-1)!}$. We factor out e^{-1} and we shift the u index, to get $E(U) = (e^{-1}) \sum_{u=0}^{\infty} \frac{1}{u!} = (e^{-1})(e^1) = 1$.

By the same reasoning, we have $E(V) = 1$ and $E(W) = 1$.

Since $X = U + V + W$, and $E(X) = E(U + V + W)$, by linearity of expectation it follows that $E(X) = 1 + 1 + 1 = 3$.